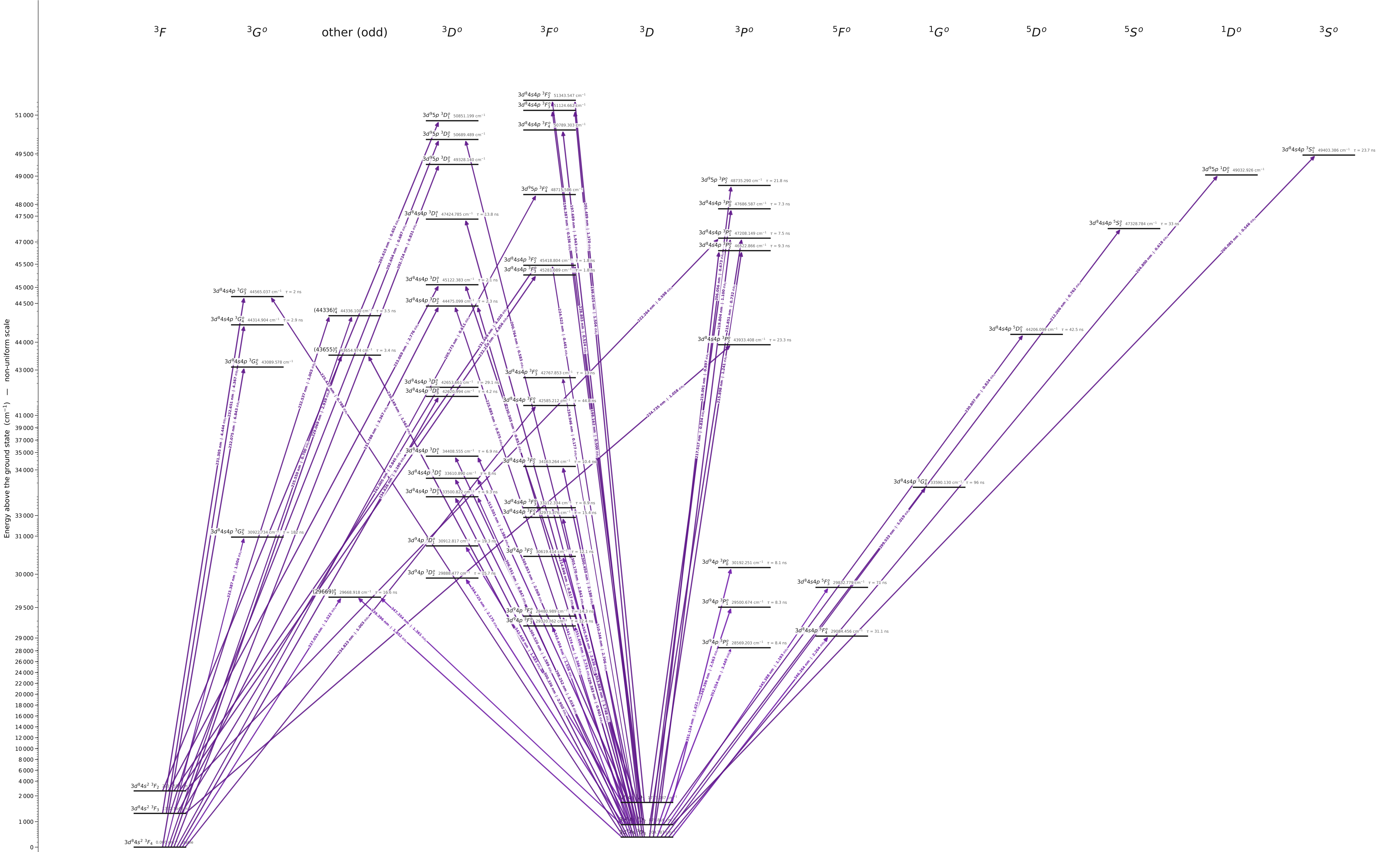
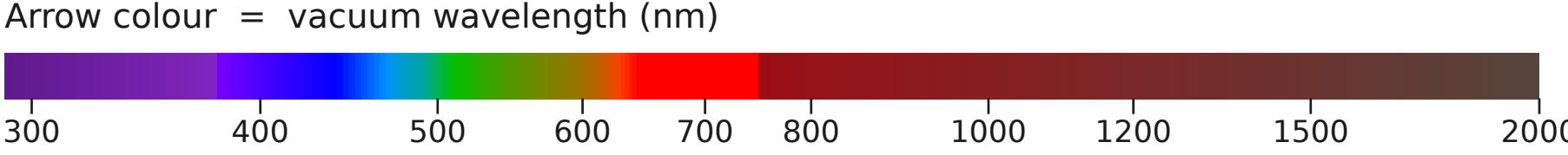


Nickel

Ni (Ni I). Strongest classified transitions below 2 μm (NIST): 55 levels, 70 lines.

Ni⁺ ionisation limit 61619.77 cm⁻¹ = 7.63988 eV (162.286 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U||e\mathbf{r}||U'\rangle$ in ea₀
Matrix elements are measured or NIST values: * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U e\mathbf{r} U'\rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
3d ⁸ 4s ² ³ F ₄ - 3d ⁸ 4s4p ³ G ₅ ^o	323.3867	323.2933	927.0402	1.004	5.490e+06	0.8745	99.9	323 nm J=4 -> J'=5 line from the 3F4 ground level

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2τ).

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
3d ⁹ 4s ² D ₃ - 3d ⁸ 4s4p ² S ₂ ^o	212.206	60-58	-800(50)
3d ⁹ 4s ² D ₂ - 3d ⁸ 4s4p ² D ₂ ^o	214.846	60-58	-710(50)
3d ⁹ 4s ² D ₂ - 3d ⁸ 4s4p ¹ P ₁ ^o	215.851	60-58	-840(50)
3d ⁹ 4s ² D ₃ - 3d ⁸ 4s4p ¹ P ₂ ^o	215.898	60-58	-770(50)
3d ⁹ 4s ² D ₁ - 3d ⁸ 4s4p ¹ P ₀ ^o	217.517	60-58	-890(40)
3d ⁹ 4s ² D ₂ - 3d ⁸ 4s4p ¹ P ₂ ^o	219.091	60-58	-750(40)
3d ⁹ 4s ² D ₁ - 3d ⁸ 4s4p ¹ P ₁ ^o	219.804	60-58	-870(40)
3d ⁹ 4s ² D ₂ - 3d ⁸ 4s4p ¹ F ₃ ^o	224.522	60-58	-1070(40)
3d ⁹ 4s ² D ₃ - 3d ⁸ 4s4p ¹ G ₅ ^o	225.427	60-58	-1060(40)
3d ⁹ 4s ² D ₃ - 3d ⁸ 4s4p ¹ D ₂ ^o	225.885	60-58	-1060(40)
3d ⁹ 4s ² D ₁ - 3d ⁸ 4s4p ¹ F ₃ ^o	228.803	60-58	-1090(40)
3d ⁸ 4s ² ³ F ₄ - (43655) ₄ ^o	229.069	60-58	740(40)
3d ⁹ 4s ² D ₂ - 3d ⁸ 4s4p ¹ D ₂ ^o	229.383	60-58	-970(40)
3d ⁹ 4s ² D ₃ - (43655) ₅ ^o	230.149	60-58	-960(40)
3d ⁹ 4s ² D ₁ - 3d ⁸ 4s4p ¹ D ₂ ^o	230.365	60-58	-1070(40)
3d ⁸ 4s ² ³ F ₃ - 3d ⁸ 4s4p ¹ G ₅ ^o	231.305	60-58	730(40)
3d ⁸ 4s ² ³ F ₂ - 3d ⁸ 4s4p ¹ F ₃ ^o	231.469	60-58	620(40)
3d ⁸ 4s ² ³ F ₃ - 3d ⁸ 4s4p ¹ D ₂ ^o	231.788	60-58	780(40)
3d ⁸ 4s ² ³ F ₂ - 3d ⁸ 4s4p ¹ F ₃ ^o	232.210	60-58	720(40)
3d ⁸ 4s ² ³ F ₃ - 3d ⁸ 4s4p ¹ G ₅ ^o	232.651	60-58	780(40)
3d ⁸ 4s ² ³ F ₂ - 3d ⁸ 4s4p ¹ D ₂ ^o	233.069	60-58	660(40)
3d ⁸ 4s ² ³ F ₄ - 3d ⁸ 4s4p ¹ D ₂ ^o	234.626	60-58	710(40)

Shift = ν(first isotope) - ν(second isotope).